

**THE MAHATMA GANDHI UNIVERSITY**  
**UNDERGRADUATE VOCATIONAL**  
**PROGRAMMES**

**(HONOURS) SYLLABUS**

**MGU-B.VOC. (Honours)**

**(2025 Admission Onwards)**



**Faculty: Science**

**Expert Committee: Agriculture & Sustainability**

**Subject: CHEMISTRY**

**Mahatma Gandhi University**  
**Priyadarshini Hills**  
**Kottayam – 686560, Kerala, India**

## Syllabus Index

Name of the Minor: **CHEMISTRY**

### Semester: 1

| Course Code  | Title of the Course                        | Type of the Course | Credit | Hours/ week | Hour Distribution /week |   |   |
|--------------|--------------------------------------------|--------------------|--------|-------------|-------------------------|---|---|
|              |                                            |                    |        |             | L                       | P | O |
| MG1MPCCHY100 | Basic Theoretical and Analytical Chemistry | MPC                | 4      | 5           | 3                       | 2 |   |

L — Lecture, P — Practical/Practicum , O — On the Job Training

### Semester: 2

| Course Code  | Title of the Course     | Type of the Course | Credit | Hours/ week | Hour Distribution /week |   |   |
|--------------|-------------------------|--------------------|--------|-------------|-------------------------|---|---|
|              |                         |                    |        |             | L                       | P | O |
| MG2MPCCHY100 | Basic Organic Chemistry | MPC                | 4      | 4           | 4                       |   |   |

L — Lecture, P — Practical/Practicum , O — On the Job Training

### Semester: 3

| Course Code  | Title of the Course   | Type of the Course | Credit | Hours/ week | Hour Distribution /week |   |   |
|--------------|-----------------------|--------------------|--------|-------------|-------------------------|---|---|
|              |                       |                    |        |             | L                       | P | O |
| MG3MPCCHY200 | Bio-Organic Chemistry | MPC                | 4      | 5           | 3                       | 2 |   |

L — Lecture, P — Practical/Practicum , O — On the Job Training

**Semester: 4**

| Course Code  | Title of the Course             | Type of the Course | Credit | Hours/<br>week | Hour Distribution<br>/week |   |   |
|--------------|---------------------------------|--------------------|--------|----------------|----------------------------|---|---|
|              |                                 |                    |        |                | L                          | P | O |
| MG4MPCCHY200 | Inorganic and Organic Chemistry | MPC                | 4      | 5              | 3                          | 2 |   |

L — Lecture, P — Practical/Practicum , O — On the Job Training

**Semester: 5**

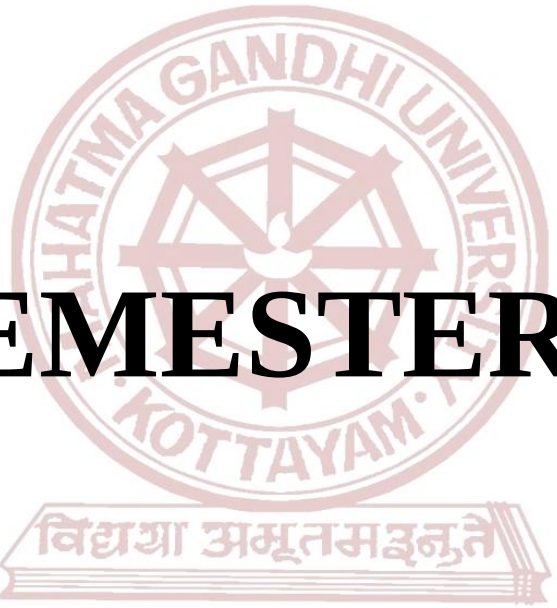
| Course Code  | Title of the Course        | Type of the Course | Credit | Hours/<br>week | Hour Distribution<br>/week |   |   |
|--------------|----------------------------|--------------------|--------|----------------|----------------------------|---|---|
|              |                            |                    |        |                | L                          | P | O |
| MG5MPCCHY300 | Chemistry in Everyday Life | MPC                | 4      | 4              | 4                          |   |   |

L — Lecture, P — Practical/Practicum , O — On the Job Training

**Syllabus Semester: 6**

| Course Code  | Title of the Course         | Type of the Course | Credit | Hours/<br>week | Hour Distribution<br>/week |   |   |
|--------------|-----------------------------|--------------------|--------|----------------|----------------------------|---|---|
|              |                             |                    |        |                | L                          | P | O |
| MG6MPCCHY300 | Advanced Physical Chemistry | MPC                | 4      | 4              | 4                          |   |   |

L — Lecture, P — Practical/Practicum , O — On the Job Training



# SEMESTER 1

MGU-B.VOC. (HONOURS)

*Syllabus*



# Mahatma Gandhi University

## Kottayam

|                        |                                                                                                                                                                                                                    |         |           |     |             |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|-----|-------------|
| Faculty/<br>Discipline | SCIENCE / CHEMISTRY                                                                                                                                                                                                |         |           |     |             |
| Programme              |                                                                                                                                                                                                                    |         |           |     |             |
| Course Name            | BASIC THEORETICAL AND ANALYTICAL CHEMISTRY                                                                                                                                                                         |         |           |     |             |
| Type of Course         | MPC                                                                                                                                                                                                                |         |           |     |             |
| Course Code            | MG1MPCCHY100                                                                                                                                                                                                       |         |           |     |             |
| Course Level           | 100-199                                                                                                                                                                                                            |         |           |     |             |
| Course Summary         | Covers the basic principles and concepts of atomic structure, chemical bonding, periodic properties and fundamental principles, techniques and calculations necessary to perform analytical chemistry experiments. |         |           |     |             |
| Semester               | 1                                                                                                                                                                                                                  | Credits |           |     | 4           |
| Course Details         | Learning Approach                                                                                                                                                                                                  | Lecture | Practical | OJT | Total Hours |
|                        |                                                                                                                                                                                                                    | 3       | 1         |     |             |
| Pre-requisites, if any | Nil                                                                                                                                                                                                                |         |           |     |             |

### COURSE OUTCOMES (CO)

| CO No. | Expected Course Outcome                                                                                              | Learning Domains | PO No |
|--------|----------------------------------------------------------------------------------------------------------------------|------------------|-------|
| 1      | Describe the basic structure of an atom and explain electronic configurations, atomic behaviour and characteristics. | K                | 1,2   |

|   |                                                                                                                                              |   |        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
| 2 | Recognize how atomic structure underpins other fundamental chemical concepts like chemical bonding, molecular structure, and reactivity.     | U | 1,2    |
| 3 | Describe and explain periodic trends such as atomic radius, ionization energy, and electronegativity.                                        | U | 1,2    |
| 4 | Understand the theoretical basis and practical applications of volumetric analysis.                                                          | U | 1,2,10 |
| 5 | Employ students with both the theoretical understanding and practical skills to accurately determine the concentration of unknown solutions. | A | 1,2,10 |

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciate (Ap).**

#### CO-PO ARTICULATION MATRIX

| CO/PO | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 |
|-------|------|------|------|------|------|------|------|------|------|-------|
| CO 1  | 3    | 3    |      |      |      |      |      |      |      |       |
| CO 2  | 3    | 3    |      |      |      |      |      |      |      |       |
| CO 3  | 3    | 3    |      |      |      |      |      |      |      |       |
| CO 4  | 3    | 3    |      |      |      |      |      |      |      | 3     |
| CO 5  | 3    | 3    |      |      |      |      |      |      |      | 3     |

**'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).**

## COURSE CONTENT

### Content for Classroom transaction (Units)

| Module | Units                                           | Course description                                                                                                                   | Hrs | CO No. |
|--------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----|--------|
| 1      | <b>ATOMIC STRUCTURE</b>                         |                                                                                                                                      |     |        |
|        | 1.1                                             | Atomic spectrum of hydrogen atom, explanation using Bohr atom model, limitations of Bohr atom model.                                 | 4   | 1      |
|        | 1.2                                             | Dual nature of matter, de Broglie equation, Heisenberg's uncertainty principle and its significance                                  | 2   | 1      |
|        | 1.3                                             | Concept of orbit and orbital. Types of orbitals, shapes of s, p and d orbitals.                                                      | 2   | 1      |
|        | 1.4                                             | Quantum numbers and their significance.                                                                                              | 2   | 1      |
|        | 1.5                                             | Pauli's Exclusion Principle, Hund's rule of maximum multiplicity and Aufbau principle.                                               | 2   | 1      |
|        | 1.6                                             | Electronic configuration of atoms (upto atomic number 30). Stability of half-filled and completely filled electronic configurations. | 3   | 1      |
| 2      | <b>CHEMICAL BONDING AND PERIODIC PROPERTIES</b> |                                                                                                                                      |     |        |
|        | 2.1                                             | Introduction- Types of bonds. Ionic bond: factors favouring the formation of ionic bonds, lattice energy.                            | 2   | 2      |
|        | 2.2                                             | covalent bond: lewis concept of covalent bond, coordinate covalent bond or dative bond.                                              | 2   | 2      |
|        | 2.3                                             | Intermolecular forces - Hydrogen bond, Vanderwaals forces.                                                                           | 2   | 2      |



|   |                                                 |                                                                                                                                                                                                                      |    |   |
|---|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|
|   | 2.4                                             | valence bond theory, geometry of water molecule, limitations of valence bond theory.                                                                                                                                 | 3  | 2 |
|   | 2.5                                             | Periodic properties: Modern periodic law-long form of periodic table.                                                                                                                                                | 2  | 3 |
|   | 2.6                                             | Periodicity in properties: Atomic radii, ionic radii, ionization enthalpy, electron affinity and electronegativity.                                                                                                  | 4  | 3 |
| 3 | <b>BASIC CONCEPTS OF ANALYTICAL CHEMISTRY</b>   |                                                                                                                                                                                                                      |    |   |
|   | 3.1                                             | Laboratory operations (non evaluative): laboratory safety and first aid. use of different glassware like pipette, burette, standard measuring flask, distillation apparatus, heating methods, filtration techniques. | 2  | 4 |
|   | 3.2                                             | Volumetric methods of analysis: general principles, primary and secondary standards, preparation of standard solutions, standardisation of solutions, indicators, endpoint.                                          | 3  | 4 |
|   | 3.3                                             | Types of titrations: acid- base, redox, precipitation and complex formation titrations.                                                                                                                              | 2  | 4 |
|   | 3.4                                             | Methods of expressing concentration: weight percentage, molarity, molality, normality, mole fraction, ppm and ppb.                                                                                                   | 5  | 4 |
|   | 3.5                                             | Acid base concepts: Arrhenius definition, Lowry - Bronsted definition and conjugate acid - base pairs, Lewis concept.                                                                                                | 3  | 4 |
| 4 | <b>PRACTICAL - ACIDIMETRY &amp; ALKALIMETRY</b> |                                                                                                                                                                                                                      |    |   |
|   | 4.1                                             | Calibration of apparatus- standard flask and preparation of standard molar solutions of                                                                                                                              | 15 | 5 |



|   |                                 |                                                                                                                                                              |    |   |
|---|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|
|   |                                 | any two primary standards-<br>oxalic acid, mohr's salt,<br>$\text{Na}_2\text{CO}_3$ .                                                                        |    |   |
|   | 4.2                             | Acid base titration- acidimetry<br>and alkalimetry: titration of<br>strong acid vs strong base,<br>strong acid vs weak base and<br>weak acid vs strong base. | 10 | 5 |
|   | 4.3                             | Estimation of chloride ion in<br>tap water.<br>Estimation of hardness in water<br>estimation of citric acid in<br>citrus fruits.                             | 5  | 5 |
| 5 | <b>Teacher Specific Content</b> |                                                                                                                                                              |    |   |
|   |                                 |                                                                                                                                                              |    |   |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Teaching and Learning Approach</b> | <b>Classroom Procedure (Mode of transaction)</b><br>Lecture Sessions, interactive sessions including discussions, demonstrations, and experiments to engage students actively and visual aids like presentations, videos, and models to enhance understanding. Encourage students to ask questions during or after the lectures. Begin with safety instructions and guidelines for lab work. Allow students to conduct experiments under supervision (for lab work). |
| <b>Assessment Types</b>               | <b>MODE OF ASSESSMENT</b><br>Mode of assessment: <b>Both</b>                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                       | <b>A. Continuous Comprehensive Assessment (CCA)</b><br><br><b>Theory</b><br>Total Marks – 25 Marks<br>Assessment Methods <ul style="list-style-type: none"> <li>• Assignments</li> <li>• MCQ</li> <li>• Viva</li> <li>• Class tests</li> </ul>                                                                                                                                                                                                                       |

### **Practical**

Total Marks – 15 Marks

Assessment Methods

- Lab skill
- Record
- Viva

### **B. End Semester Evaluation (ESE)**

#### **Theory**

Total Marks – 50 Marks

**Duration of Examination:** 1.5 Hours

**Pattern of examination for Theory:** Non - MCQ

**Different parts of written examination:** Part A, Part B, Part C

**Answer Type:** One sentence, short answer, essay

**Marks for part:**

- Part A  
Answer in a sentence - (10 out of 12) -  $10 \times 1 = 10$
- Part B  
Short answer- (4 out of 6) -  $4 \times 5 = 20$
- Part C  
Essay - (2 out of 4) -  $2 \times 10 = 20$

#### **Practical**

Total Marks: 35 Marks

Assessment Methods:

- Lab report :10 marks
- Viva: 10 marks
- Writing procedure and calculation and accuracy:15 marks

**Duration of Examination:** 3 Hours

## REFERENCES

- P. W. Atkins and J. de Paula, Physical Chemistry, 11th Edn. Oxford University Press, 2018.
- Vogel's Textbook of Quantitative Chemical Analysis, 6th Edn. Pearson Education Ltd.
- T. Brown, C. Murphy, H. LeMay, Laboratory Experiments for Chemistry, Pearson, 2018.
- B.R. Puri, L.R. Sharma, M.S. Pathania, Elements of Physical Chemistry, 3rd edn. Vishal Pub. Co., 2008.
- C.N.R. Rao, University General Chemistry, Macmillan, 2009.
- Manas Chanda, Atomic Structure and Molecular Spectroscopy.



**MGU-B.VOC. (HONOURS)**

## Syllabus

The logo of Mahatma Gandhi University, Kottayam, is a circular emblem. It features a central geometric design resembling a stylized 'G' or a lotus flower. The text 'MAHATMA GANDHI UNIVERSITY' is written in a circle around the top, and 'KOTTAYAM' is at the bottom. Below the circle is a banner with the Sanskrit motto 'विद्यया अमृतमश्नुते' (Vidya Ya Amritam Ashnute).

# SEMESTER 2

**MGU-B.VOC. (HONOURS)**

*Syllabus*



# Mahatma Gandhi University

## Kottayam

|                        |                                                                                                                                                                            |         |           |     |             |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|-----|-------------|
| Faculty/<br>Discipline | SCIENCE / CHEMISTRY                                                                                                                                                        |         |           |     |             |
| Programme              |                                                                                                                                                                            |         |           |     |             |
| Course Name            | BASIC ORGANIC CHEMISTRY                                                                                                                                                    |         |           |     |             |
| Type of Course         | MPC                                                                                                                                                                        |         |           |     |             |
| Course Code            | MG2MPCCHY100                                                                                                                                                               |         |           |     |             |
| Course Level           | 100-199                                                                                                                                                                    |         |           |     |             |
| Course Summary         | Explores electron displacements in organic chemistry, reactive intermediates, mechanism of organic reactions, basic stereochemistry and fundamentals of polymer chemistry. |         |           |     |             |
| Semester               | 2                                                                                                                                                                          | Credits |           | 4   | Total Hours |
| Course Details         | Learning Approach                                                                                                                                                          | Lecture | Practical | OJT |             |
|                        |                                                                                                                                                                            | 4       |           |     |             |
| Pre-requisites, if any | Nil                                                                                                                                                                        |         |           |     |             |

### COURSE OUTCOMES (CO)

| CO No. | Expected Course Outcome                                                                          | Learning Domains | PO No  |
|--------|--------------------------------------------------------------------------------------------------|------------------|--------|
| 1      | Define arrow formalism, reactive intermediates, electron displacement effects and hybridisation. | K                | 1,2,10 |

|   |                                                                              |   |            |
|---|------------------------------------------------------------------------------|---|------------|
| 2 | Describe the mechanisms of organic reactions and types of organic reactions. | U | 1,2        |
| 3 | Explain the basic stereochemistry of organic compounds.                      | U | 1,2        |
| 4 | Describe the fundamentals of polymer chemistry.                              | U | 1,2,6,7,10 |

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciate (Ap).**

#### CO-PO ARTICULATION MATRIX

| CO/PO | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 |
|-------|------|------|------|------|------|------|------|------|------|-------|
| CO 1  | 3    | 3    |      |      |      |      |      |      |      |       |
| CO 2  | 2    | 3    |      |      |      |      |      |      |      |       |
| CO 3  | 2    | 3    |      |      |      |      |      |      |      |       |
| CO 4  | 3    | 3    |      |      |      | 2    | 2    |      |      | 2     |

*'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).*

#### COURSE CONTENT

| Module | Units                                    | Course description                                                                         | Hrs | CO No. |
|--------|------------------------------------------|--------------------------------------------------------------------------------------------|-----|--------|
| 1      | <b>FUNDAMENTALS OF ORGANIC CHEMISTRY</b> |                                                                                            |     |        |
|        | 1.1                                      | Arrow formalism in organic chemistry, bond fission                                         | 2   | 1      |
|        | 1.2                                      | Types of reagents: nucleophiles, electrophiles.                                            | 2   | 1      |
|        | 1.3                                      | Reaction intermediates or reactive intermediates: carbocations, carbanions, free radicals. | 4   | 1      |
|        | 1.4                                      | Electron displacement effects: inductive effect, mesomeric effect, hyperconjugation.       | 2   | 1      |

|   |                                                   |                                                                                                                     |   |   |
|---|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---|---|
|   | 1.5                                               | Hybridisation- sp <sup>3</sup> , sp <sup>2</sup> and sp.<br>(ethane, ethene and ethyne)                             | 5 | 1 |
| 2 | <b>MECHANISM OF ORGANIC REACTIONS</b>             |                                                                                                                     |   |   |
|   | 2.1                                               | meaning of reaction mechanism, polarity of bonds.                                                                   | 1 | 2 |
|   | 2.2                                               | Types of organic reactions: substitution, addition, elimination, rearrangement and redox reactions.                 | 2 | 2 |
|   | 2.3                                               | substitution reactions: nucleophilic substitution of alkyl halides- SN <sub>1</sub> and SN <sub>2</sub> mechanisms. | 4 | 2 |
|   | 2.4                                               | Electrophilic substitution in benzene.                                                                              | 2 | 2 |
|   | 2.5                                               | Addition reactions: electrophilic addition to alkenes and alkynes, Markwonikoff's rule, Peroxide effect.            | 4 | 2 |
|   | 2.6                                               | Elimination reactions: E1 and E2 mechanisms. Saytzeff's rule and Hofmann's rule.                                    | 2 | 2 |
| 3 | <b>BASIC STEREOCHEMISTRY OF ORGANIC COMPOUNDS</b> |                                                                                                                     |   |   |
|   | 3.1                                               | Stereoisomerism , geometrical isomerism                                                                             | 2 | 3 |
|   | 3.2                                               | E and Z nomenclature of geometrical isomers                                                                         | 2 | 3 |
|   | 3.3                                               | Optical isomerism: D and L configurations, optical activity, chirality, chiral centre.                              | 3 | 3 |
|   | 3.4                                               | Enantiomers, racemic modification, resolution, Diastereomers, Meso form.                                            | 4 | 3 |
|   | 3.5                                               | Conformation; conformation of ethane, butane and cyclohexane.                                                       | 4 | 3 |
| 4 | <b>INTRODUCTION TO POLYMER CHEMISTRY</b>          |                                                                                                                     |   |   |
|   | 4.1                                               | Classification of polymers; Natural and synthetic, organic and inorganic,                                           | 3 | 4 |



|   |                                 |                                                                                                      |   |   |
|---|---------------------------------|------------------------------------------------------------------------------------------------------|---|---|
|   |                                 | thermoplastic and thermosetting, plastics, elastomers and fibres.                                    |   |   |
|   | 4.2                             | Linear, branched and crosslinked polymers.<br>Homopolymers and Heteropolymers.                       | 2 | 4 |
|   | 4.3                             | Polymerisation reactions: addition and condensation polymerisation.                                  | 2 | 4 |
|   | 4.4                             | Plastics: polyethylene, polypropylene, PVC, Phenol formaldehyde resin, MF resin, Nylons, polyesters. | 5 | 4 |
|   | 4.5                             | Natural rubber, synthetic rubber.<br>Biodegradability of polymers, environmental hazards.            | 3 | 4 |
| 5 | <b>Teacher Specific Content</b> |                                                                                                      |   |   |
|   |                                 |                                                                                                      |   |   |

### Content for Classroom transaction (Units)

|                                       |                                                                                                                                                                                                                                                                    |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Teaching and Learning Approach</b> | <b>Classroom Procedure (Mode of transaction)</b><br><br>Lecture sessions, interactive sessions including discussions, demonstrations, and experiments to engage students actively and visual aids like presentations, videos, and models to enhance understanding. |
| <b>Assessment Types</b>               | <b>MODE OF ASSESSMENT</b><br><br>Mode of assessment: <b>Theory</b>                                                                                                                                                                                                 |
|                                       | <b>A. Continuous Comprehensive Assessment (CCA)</b><br><br><b>Theory</b><br><br>Total Marks – 30 Marks<br><br>Assessment Methods                                                                                                                                   |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <ul style="list-style-type: none"> <li>• Assignments</li> <li>• MCQ</li> <li>• Viva</li> <li>• Class tests</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|  | <p><b>End Semester Evaluation (ESE)</b></p> <p><b>Theory</b></p> <p>Total Marks – 70 Marks</p> <p><b>Duration of Examination:</b> 2 Hours</p> <p><b>Pattern of examination for Theory:</b> Non - MCQ</p> <p><b>Different parts of written examination:</b> Part A, Part B, Part C</p> <p><b>Answer Type:</b> Two sentence, short answer, essay</p> <p><b>Marks for part:</b></p> <ul style="list-style-type: none"> <li>• Part A<br/>Answer in two sentences - (10 out of 12) - <math>10 \times 2 = 20</math></li> <li>• Part B<br/>Short answer- (6 out of 8) - <math>6 \times 5 = 30</math></li> <li>• Part C<br/>Essay - (2 out of 4) - <math>2 \times 10 = 20</math></li> </ul> |

## REFERENCES

## MGU-B.VOC. (HONOURS)

- L. Finar, Organic Chemistry Vol. I, 6th edn. Pearson(Chapter 2-4)
- Bahl, A.; Bahl, B. S. A Textbook of Organic Chemistry; S. Chand, 2010.
- Jain, M. K.; Sharma, S. C. Modern Organic Chemistry; Vishal Publishing Co., 2010.
- McMurry, J. E. Fundamentals of Organic Chemistry; Cengage Learning, 2010.
- Clayden, J.; Greeves, N.; Warren, S. Organic Chemistry; Oxford University Press, USA, 2012.
- Wade, L. G. Organic Chemistry; Pearson Education India, 2008.
- Bruice, P. Y. Organic Chemistry; Pearson, 2017.

# SEMESTER 3



MGU-B.VOC. (HONOURS)

*Syllabus*



# Mahatma Gandhi University

## Kottayam

|                                   |                                                                                           |                |                  |                    |
|-----------------------------------|-------------------------------------------------------------------------------------------|----------------|------------------|--------------------|
| <b>Faculty/<br/>Discipline</b>    | <b>SCIENCE / CHEMISTRY</b>                                                                |                |                  |                    |
| <b>Programme</b>                  |                                                                                           |                |                  |                    |
| <b>Course Name</b>                | <b>BIO ORGANIC CHEMISTRY</b>                                                              |                |                  |                    |
| <b>Type of<br/>Course</b>         | MPC                                                                                       |                |                  |                    |
| <b>Course Code</b>                | <b>MG3MPCCHY200</b>                                                                       |                |                  |                    |
| <b>Course Level</b>               | <b>200-299</b>                                                                            |                |                  |                    |
| <b>Course<br/>Summary</b>         | Delve to the fundamentals of biochemistry, enzymes, nucleic acids. vitamins and minerals. |                |                  |                    |
| <b>Semester</b>                   | 3                                                                                         | <b>Credits</b> |                  | 4                  |
| <b>Course Details</b>             | <b>Learning<br/>Approach</b>                                                              | <b>Lecture</b> | <b>Practical</b> | <b>OJT</b>         |
|                                   |                                                                                           | 3              | 1                |                    |
|                                   |                                                                                           |                |                  | <b>Total Hours</b> |
|                                   |                                                                                           |                |                  | 75                 |
| <b>Pre-requisites,<br/>if any</b> | Nil                                                                                       |                |                  |                    |

### COURSE OUTCOMES (CO)

| CO No. | Expected Course Outcome                                                    | Learning Domains | PO No |
|--------|----------------------------------------------------------------------------|------------------|-------|
| 1      | Explain the concepts of terpenes, amino acids, proteins and carbohydrates. | U                | 1,2,3 |

|   |                                                                                                     |   |        |
|---|-----------------------------------------------------------------------------------------------------|---|--------|
| 2 | Understand the concepts of enzyme action and kinetics of enzyme catalysis.                          | U | 1,2    |
| 3 | Explain structure and functions of Nucleic acids.                                                   | U | 1,2    |
| 4 | Describe minerals, distinguish fortification and enrichment                                         | U | 1,2,7  |
| 5 | Produce proficiency in qualitative analysis of carbohydrates, proteins and in analysis of minerals. | A | 1,2,10 |

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciate (Ap).**

#### CO-PO ARTICULATION MATRIX

| CO/PO | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 |
|-------|------|------|------|------|------|------|------|------|------|-------|
| CO 1  | 3    | 3    | 1    |      |      |      |      |      |      |       |
| CO 2  | 2    | 3    |      |      |      |      |      |      |      |       |
| CO 3  | 2    | 3    |      |      |      |      |      |      |      |       |
| CO 4  | 2    | 3    |      |      |      |      | 1    |      |      |       |
| CO 5  | 3    | 3    |      |      |      |      |      |      |      | 2     |

**'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).**

**MGU-B.VOC. (HONOURS)**

#### COURSE CONTENT

**Content for Classroom transaction (Units)**

| Module | Units                               | Course description                                                                             | Hrs | CO No. |
|--------|-------------------------------------|------------------------------------------------------------------------------------------------|-----|--------|
| 1      | <b>INTRODUCTION TO BIOCHEMISTRY</b> |                                                                                                |     |        |
|        | 1.1                                 | Terpenes- Essential oils and alkaloids (structure, isolation and properties - elementary idea) | 3   | 1      |
|        | 1.2                                 | Lipids- classification and properties (elementary idea)                                        | 2   | 1      |

|   |                                  |                                                                                                                          |   |   |
|---|----------------------------------|--------------------------------------------------------------------------------------------------------------------------|---|---|
|   | 1.3                              | Amino acids; classification, separation and properties.                                                                  | 3 | 1 |
|   | 1.4                              | Proteins: classification, properties, structure.                                                                         | 5 | 1 |
|   | 1.5                              | Carbohydrates: classification and examples.                                                                              | 2 | 1 |
| 2 | <b>ENZYMES AND NUCLEIC ACIDS</b> |                                                                                                                          |   |   |
|   | 2.1                              | Nomenclature and classification, characteristics of enzymes.                                                             | 2 | 2 |
|   | 2.2                              | Mechanism of enzyme action, kinetics of enzyme catalysis, factors affecting enzyme action.                               | 4 | 2 |
|   | 2.3                              | Cofactors - coenzymes, enzyme inhibitors, enzyme deficiency diseases, applications of enzymes in the food industry.      | 3 | 2 |
|   | 2.4                              | Energy rich molecules- ATP and ADP.                                                                                      | 2 | 3 |
|   | 2.5                              | Nucleic acids: hydrolysis and structure.                                                                                 | 2 | 3 |
|   | 2.6                              | Functions of nucleic acids; replication of DNA, protein biosynthesis.                                                    | 2 | 3 |
| 3 | <b>VITAMINS AND MINERALS</b>     |                                                                                                                          |   |   |
|   | 3.1                              | Vitamins: classification with examples. Structure, biological functions and deficiency diseases of Fat-soluble vitamins. | 5 | 4 |
|   | 3.2                              | water soluble vitamins - structure, functions and deficiency diseases.                                                   | 5 | 4 |
|   | 3.3                              | Classification of minerals, Fortification, enrichment and restoration.                                                   | 5 | 4 |

|   |                                 |                                                                       |    |   |
|---|---------------------------------|-----------------------------------------------------------------------|----|---|
| 4 | <b>PRACTICAL</b>                |                                                                       |    |   |
|   | 4.1                             | Analysis of sodium, potassium and calcium by flame photometer method. | 15 | 5 |
|   | 4.2                             | Qualitative analysis of carbohydrates, proteins                       | 10 | 5 |
|   | 4.3                             | Estimation of vitamins                                                | 5  | 5 |
| 5 | <b>Teacher Specific Content</b> |                                                                       |    |   |
|   |                                 |                                                                       |    |   |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Teaching and Learning Approach</b> | <b>Classroom Procedure (Mode of transaction)</b><br><br>Lecture Sessions, interactive sessions including discussions, demonstrations, and experiments to engage students actively and visual aids like presentations, videos, and models to enhance understanding. Begin with safety instructions and guidelines for lab work. Allow students to conduct experiments under supervision (for lab work). |
| <b>Assessment Types</b>               | <b>MODE OF ASSESSMENT</b><br><br>Mode of assessment: <b>Both</b>                                                                                                                                                                                                                                                                                                                                       |
|                                       | <b>A. Continuous Comprehensive Assessment (CCA)</b><br><br><b>Theory</b><br><br>Total Marks – 25 Marks<br><br>Assessment Methods <ul style="list-style-type: none"> <li>● Assignments</li> <li>● MCQ</li> <li>● Viva</li> <li>● Class tests</li> </ul>                                                                                                                                                 |



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>Practical</b></p> <p>Total Marks – 15 Marks</p> <p>Assessment Methods</p> <ul style="list-style-type: none"> <li>• Lab skill</li> <li>• Record</li> <li>• Viva</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|  | <p><b>A. End Semester Evaluation (ESE)</b></p> <p><b>Theory</b></p> <p>Total Marks – 50 Marks</p> <p><b>Duration of Examination:</b> 1.5 Hours</p> <p><b>Pattern of examination for Theory:</b> Non - MCQ</p> <p><b>Different parts of written examination:</b> Part A, Part B, Part C</p> <p><b>Answer Type:</b> One sentence, short answer, essay</p> <p><b>Marks for part:</b></p> <ul style="list-style-type: none"> <li>• Part A<br/>Answer in a sentence - (10 out of 12) - <math>10 \times 1 = 10</math></li> <li>• Part B<br/>Short answer- (4 out of 6) - <math>4 \times 5 = 20</math></li> <li>• Part C<br/>Essay - (2 out of 4) - <math>2 \times 10 = 20</math></li> </ul> <p><b>Practical</b></p> <p>Total Marks: 35 Marks</p> <p>Assessment Methods:</p> <ul style="list-style-type: none"> <li>• Lab report :10 marks</li> <li>• Viva: 10 marks</li> <li>• Writing procedure, calculation and accuracy:15 marks</li> </ul> <p><b>Duration of Examination:</b> 3 Hours</p> |

## REFERENCES

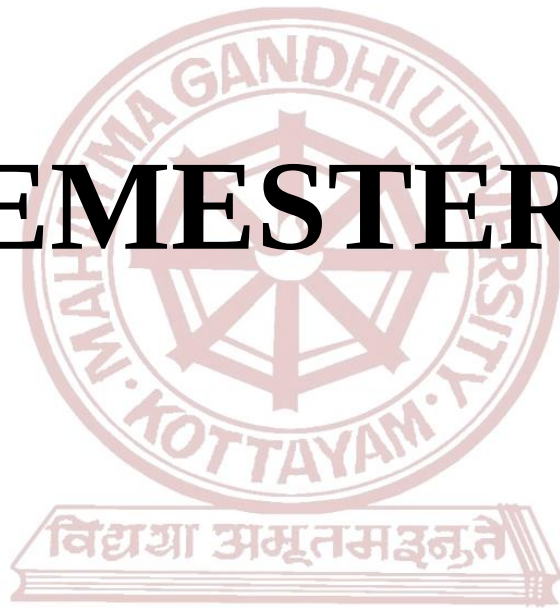
- S.Suzanne Nielsen, Food Analysis,3rd Edn. Aspen Publication
- Lehninger Principles of Biochemistry by David L. Nelson and Michael M. Cox:
- Fennema's Food Chemistry by Srinivas Vadivambal and Gary R. List (Editors) / Owen R. Fennema
- The Science of Food: An Introduction to Food Science, Nutrition and Microbiology by P. M. Gaman and K. B. Sherrington:



**MGU-B.VOC. (HONOURS)**

## Syllabus

# SEMESTER 4



**MGU-B.VOC. (HONOURS)**

*Syllabus*



# Mahatma Gandhi University

## Kottayam

|                        |                                                                                                                                                                        |         |           |     |             |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|-----|-------------|
| Faculty/<br>Discipline | SCIENCE / CHEMISTRY                                                                                                                                                    |         |           |     |             |
| Programme              |                                                                                                                                                                        |         |           |     |             |
| Course Name            | INORGANIC AND ORGANIC CHEMISTRY                                                                                                                                        |         |           |     |             |
| Type of Course         | MPC                                                                                                                                                                    |         |           |     |             |
| Course Code            | MG4MPCCHY200                                                                                                                                                           |         |           |     |             |
| Course Level           | 200-299                                                                                                                                                                |         |           |     |             |
| Course Summary         | To apprehend the chemistry of living cells, differentiate natural and synthetic food pigments and colours and to explain the core concept of supramolecular chemistry. |         |           |     |             |
| Semester               | 4                                                                                                                                                                      | Credits |           | 4   | Total Hours |
| Course Details         | Learning Approach                                                                                                                                                      | Lecture | Practical | OJT |             |
|                        |                                                                                                                                                                        | 3       | 1         |     |             |
| Pre-requisites, if any | Nil                                                                                                                                                                    |         |           |     |             |

### COURSE OUTCOMES (CO)

| CO No. | Expected Course Outcome                              | Learning Domains | PO No  |
|--------|------------------------------------------------------|------------------|--------|
| 1      | Describe molecular structure to biological function. | U                | 1,2,10 |

|   |                                                                        |   |          |
|---|------------------------------------------------------------------------|---|----------|
| 2 | Differentiate natural and synthetic food colours and pigments.         | U | 1,2,7,10 |
| 3 | Identify and describe non covalent interactions.                       | U | 1,2      |
| 4 | Estimate chlorophyll, carotenoid and flavonoid content in the samples. | A | 1,2,7    |

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciate (Ap).**

#### CO-PO ARTICULATION MATRIX

| CO/PO | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 |
|-------|------|------|------|------|------|------|------|------|------|-------|
| CO 1  | 2    | 3    |      |      |      |      |      |      |      | 2     |
| CO 2  | 2    | 3    |      |      |      |      | 2    |      |      | 2     |
| CO 3  | 3    | 3    |      |      |      |      |      |      |      |       |
| CO 4  | 2    | 3    |      |      |      |      | 1    |      |      |       |

*'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).*

#### COURSE CONTENT

##### Content for Classroom transaction (Units)

| Module | Units                           | Course description                                                                                                                             | Hrs | CO No. |
|--------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|
| 1      | <b>CHEMISTRY OF LIVING CELL</b> |                                                                                                                                                |     |        |
|        | 1.1                             | Thermodynamics of living cells - exergonic and endergonic reactions.                                                                           | 2   | 1      |
|        | 1.2                             | Biochemistry of iron-metalloporphyrins, haemoglobin, myoglobin, pH of blood, bohr effect, oxygen dissociation curve, cytochromes, ferredoxins. | 5   | 1      |
|        | 1.3                             | Photosynthesis - photosystems, photophosphorylation                                                                                            | 4   | 1      |

|   |                                  |                                                                                                                                                                               |    |   |
|---|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|
|   | 1.4                              | Sodium- potassium pump.<br>Biochemistry of zinc and cobalt.                                                                                                                   | 4  | 1 |
| 2 | <b>FOOD COLOURS AND PIGMENTS</b> |                                                                                                                                                                               |    |   |
|   | 2.1                              | Introduction to colours and pigments in food.                                                                                                                                 | 2  | 2 |
|   | 2.2                              | Natural food pigments-<br>Introduction to natural pigments, major classes of natural pigments: chlorophylls, carotenoids, anthocyanins, flavonoids, betalains, heme pigments. | 7  | 2 |
|   | 2.3                              | Synthetic food colours-<br>Introduction, major classes of synthetic dyes: azo dyes, triphenylmethane dyes, xanthene dyes, indigoid dyes.                                      | 6  | 2 |
| 3 | <b>SUPRAMOLECULAR CHEMISTRY</b>  |                                                                                                                                                                               |    |   |
|   | 3.1                              | Supramolecular chemistry-<br>definition, Host and Guest                                                                                                                       | 2  | 3 |
|   | 3.2                              | Molecular recognition, Host guest complex, molecular receptors, supermolecules.                                                                                               | 5  | 3 |
|   | 3.3                              | Host guest interactions- types of non covalent interactions.                                                                                                                  | 4  | 3 |
|   | 3.4                              | Role of molecular recognition in biopolymer( DNA and RNA) structure organisation. (elementary idea only)                                                                      | 4  | 3 |
| 4 | <b>PRACTICAL</b>                 |                                                                                                                                                                               |    |   |
|   | 4.1                              | Estimation of chlorophyll and carotenoids                                                                                                                                     | 10 | 4 |
|   | 4.2                              | Estimation of Flavonoids                                                                                                                                                      | 5  | 4 |
|   | 4.3                              | Estimation of antioxidants.                                                                                                                                                   | 5  | 4 |

|   |                                 |                                                            |    |   |
|---|---------------------------------|------------------------------------------------------------|----|---|
|   | 4.4                             | Identification of food colours using paper chromatography. | 10 | 4 |
| 5 | <b>Teacher Specific Content</b> |                                                            |    |   |
|   |                                 |                                                            |    |   |

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Teaching and Learning Approach</b> | <b>Classroom Procedure (Mode of transaction)</b><br><br>Lecture Sessions, interactive sessions including discussions, demonstrations, and experiments to engage students actively and visual aids like presentations, videos, and models to enhance understanding. Begin with safety instructions and guidelines for lab work. Allow students to conduct experiments under supervision (for lab work).                                                                                                                                                                                                        |
| <b>Assessment Types</b>               | <b>MODE OF ASSESSMENT</b><br><br>Mode of Assessment: <b>Both</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                       | <b>A. Continuous Comprehensive Assessment (CCA)</b><br><br><div style="text-align: center;"> <br/> <b>Theory</b><br/> Total Marks – 25 Marks<br/> Assessment Methods </div> <ul style="list-style-type: none"> <li>● Assignments</li> <li>● MCQ</li> <li>● viva</li> <li>● class tests</li> </ul> <div style="text-align: center;"> <b>Practical</b><br/><br/> Total Marks – 15 Marks<br/><br/> Assessment Methods </div> <ul style="list-style-type: none"> <li>● lab skill</li> <li>● record</li> <li>● viva</li> </ul> |



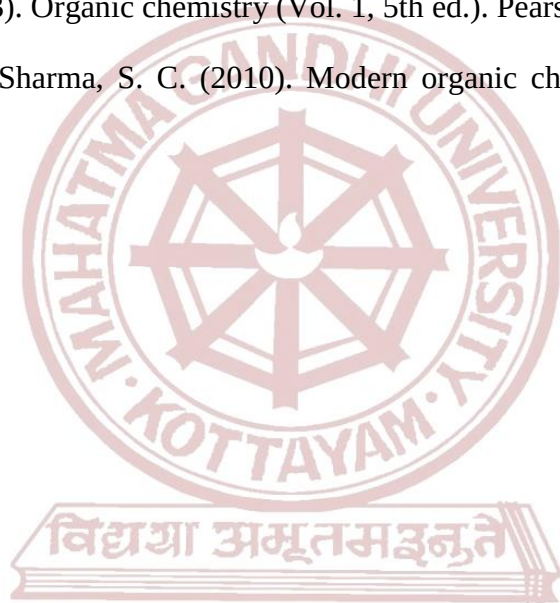
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>A. End Semester Evaluation (ESE)</b></p> <p><b>Theory</b></p> <p>Total Marks – 50 Marks</p> <p><b>Assessment Methods</b></p> <p><b>Duration of Examination:</b> 1.5 Hours</p> <p><b>Pattern of examination for Theory:</b> Non - MCQ</p> <p><b>Different parts of written examination:</b> Part A, Part B, Part C</p> <p><b>Answer Type:</b> One sentence, short answer, essay</p> <p><b>Marks for part:</b></p> <ul style="list-style-type: none"> <li>• Part A<br/>Answer in a sentence - (10 out of 12) - <math>10 \times 1 = 10</math></li> <li>• Part B<br/>Short answer- (4 out of 6) - <math>4 \times 5 = 20</math></li> <li>• Part C<br/>Essay - (2 out of 4) - <math>2 \times 10 = 20</math></li> </ul> <p><b>Practical</b></p> <p>Total Marks: 35 Marks</p> <p>Assessment Methods:</p> <ul style="list-style-type: none"> <li>• lab report :10 marks</li> <li>• viva: 10 marks</li> <li>• Writing procedure and calculation:15 marks</li> </ul> <p><b>Duration of Examination:</b> 3 Hours</p> |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## REFERENCES

- Arnikaar, H. J. (1995). Essentials of nuclear chemistry (Rev. 4th ed.). New Age.
- Coultate, T. P. (n.d.). Food: The chemistry of its components. Royal Society of Chemistry.
- Puri, B. R., Sharma, L. R., & Pathania, M. S. (2008). Elements of physical chemistry

(3rd ed.). Vishal Publishing Co.

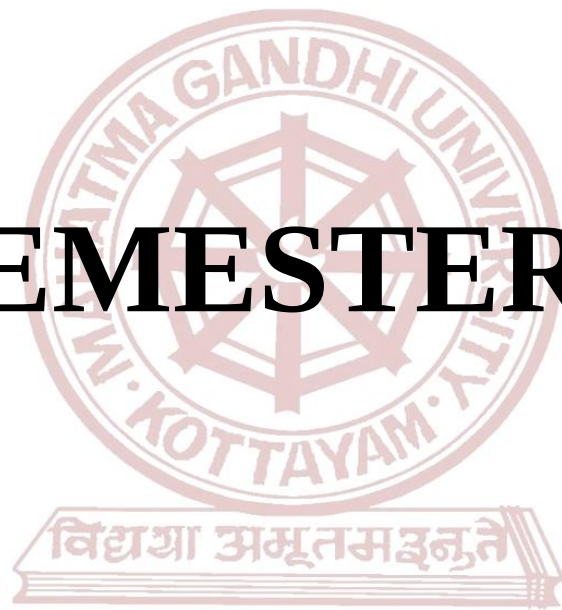
- Finar, I. L. (2002). Organic chemistry (Vols. 1–2, 6th ed.). Pearson.
- Rao, C. N. R. (2009). University general chemistry. Macmillan.
- Puri, B. R., Sharma, L. R., & Kalia, K. C. (2013). Principles of inorganic chemistry. Milestone Publishers.
- Puri, B. R., & Sharma, L. R. (n.d.). Advanced organic chemistry.
- Morrison, R. T., & Boyd, R. N. (2013). Organic chemistry (7th ed.). Pearson Education.
- Finar, I. L. (2013). Organic chemistry (Vol. 1, 5th ed.). Pearson Education.
- Jain, M. K., & Sharma, S. C. (2010). Modern organic chemistry (3rd ed.). Vishal Publishing Co.



**MGU-B.VOC. (HONOURS)**

## Syllabus

# SEMESTER 5



MGU-B.VOC. (HONOURS)

*Syllabus*



# Mahatma Gandhi University

## Kottayam

|                        |                                                                                           |         |           |     |             |
|------------------------|-------------------------------------------------------------------------------------------|---------|-----------|-----|-------------|
| Faculty/<br>Discipline | SCIENCE / CHEMISTRY                                                                       |         |           |     |             |
| Programme              |                                                                                           |         |           |     |             |
| Course Name            | CHEMISTRY IN EVERYDAY LIFE                                                                |         |           |     |             |
| Type of Course         | MPC                                                                                       |         |           |     |             |
| Course Code            | MG5MPCCHY300                                                                              |         |           |     |             |
| Course Level           | 300-399                                                                                   |         |           |     |             |
| Course Summary         | Comprehensive understanding of how chemistry permeates various aspects in our daily life. |         |           |     |             |
| Semester               | 5                                                                                         | Credits |           | 4   | Total Hours |
| Course Details         | Learning Approach                                                                         | Lecture | Practical | OJT |             |
|                        |                                                                                           | 4       |           |     |             |
| Pre-requisites, if any | Nil                                                                                       |         |           |     |             |

### COURSE OUTCOMES (CO)

| CO No. | Expected Course Outcome                            | Learning Domains | PO No      |
|--------|----------------------------------------------------|------------------|------------|
| 1      | Classify soaps and understand its cleansing action | U                | 1,2,6,7,10 |

|   |                                                                                     |   |            |
|---|-------------------------------------------------------------------------------------|---|------------|
| 2 | Understand the chemical components in personal care products.                       | U | 1,2,6,7,10 |
| 3 | Classify various types of drugs.                                                    | U | 1,2,6,7,10 |
| 4 | Explain the uses of fertilizers and pesticides and their impact on the environment. | U | 1,2,6,7,10 |
| 5 | Awareness about various types of environmental pollution.                           | U | 1,2,6,7,10 |

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciate (Ap).**

#### CO-PO ARTICULATION MATRIX

| CO/PO | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 |
|-------|------|------|------|------|------|------|------|------|------|-------|
| CO 1  | 2    | 3    |      |      |      | 2    | 3    |      |      | 2     |
| CO 2  | 2    | 3    |      |      |      | 2    | 3    |      |      | 2     |
| CO 3  | 2    | 3    |      |      |      | 2    | 3    |      |      | 2     |
| CO 4  | 2    | 3    |      |      |      | 2    | 3    |      |      | 2     |
| CO 5  | 2    | 3    |      |      |      | 2    | 3    |      |      | 2     |

**'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).**

#### COURSE CONTENT

##### Content for Classroom transaction (Units)

| Module | Units                          | Course description                                                     | Hrs | CO No. |
|--------|--------------------------------|------------------------------------------------------------------------|-----|--------|
| 1      | <b>CHEMISTRY AND COSMETICS</b> |                                                                        |     |        |
|        | 1.1                            | Soaps and detergents-<br>cleansing action and<br>environmental aspects | 5   | 1      |
|        | 1.2                            | Toothpaste- Ingredients and<br>functions                               | 3   | 2      |

|   |                                  |                                                                                                                        |   |   |
|---|----------------------------------|------------------------------------------------------------------------------------------------------------------------|---|---|
|   |                                  | Shampoos - ingredients and functions<br>Hair dyes- ingredients and functions                                           |   |   |
|   | 1.3                              | Skin products- creams and lotions,-lipstick- perfumes,colognes and aftershave lotions- deodorants and antiperspirants. | 4 | 2 |
|   | 1.4                              | Shaving cream- ingredients and functions<br>Talcum powder, Toxicology of cosmetics.                                    | 3 | 2 |
| 2 | <b>CHEMISTRY AND MEDICINE</b>    |                                                                                                                        |   |   |
|   | 2.1                              | Drugs-classification.<br>Antibiotics- classification, penicillin,ampicillin.                                           | 3 | 3 |
|   | 2.2                              | Sulpha drugs- definition, mode of action, sulphanilamide.                                                              | 3 | 3 |
|   | 2.3                              | Antipyretics and analgesics- aspirin, paracetamol                                                                      | 2 | 3 |
|   | 2.4                              | Antacids- ranitidine, antimalarials- chloroquine, anti cancer drugs, tranquilizers, antidepressants and stimulants.    | 5 | 3 |
|   | 2.5                              | Drug addiction and abuse - prevention and treatment.                                                                   | 2 | 3 |
| 3 | <b>CHEMISTRY AND AGRICULTURE</b> |                                                                                                                        |   |   |
|   | 3.1                              | Fertilizers: NPK, superphosphates, triple superphosphate, uses of mixed fertilizers.                                   | 3 | 4 |
|   | 3.2                              | Micronutrients and their role, bio- fertilizers, plant growth hormones.                                                | 2 | 4 |
|   | 3.3                              | Pesticides: classification with examples, biopesticides.                                                               | 2 | 4 |

|   |                                  |                                                                                                                                                 |   |   |
|---|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
|   | 3.4                              | Insecticides- stomach poisons, contact insecticides, fumigants. Method of preparation and use of DDT, BHC                                       | 4 | 4 |
|   | 3.5                              | Herbicides and fungicides. Excessive use of pesticides- environmental hazard.                                                                   | 4 | 4 |
|   | <b>POLLUTION AND ITS CONTROL</b> |                                                                                                                                                 |   |   |
| 4 | 4.1                              | Air pollution- causes and effects, acid rain, smog, greenhouse effect, global warming, ozone layer depletion, measures to reduce air pollution. | 5 | 5 |
|   | 4.2                              | Water pollution- causes, inorganic and macroscopic contaminants, effects of pesticides, insecticides and detergents on water pollution.         | 4 | 5 |
|   | 4.3                              | Water quality parameters - DO, BOD, COD. Measures against water pollution                                                                       | 4 | 5 |
|   | 4.4                              | Soil pollution- causes, effects and control measures.                                                                                           | 1 | 5 |
|   | 4.5                              | Noise pollution- causes, effects and control measures.                                                                                          | 1 | 5 |
| 5 | <b>Teacher Specific Content</b>  |                                                                                                                                                 |   |   |
|   |                                  |                                                                                                                                                 |   |   |



|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Teaching and Learning Approach</b> | <b>Classroom Procedure (Mode of transaction)</b><br><br>Lecture Sessions, interactive sessions including discussions, demonstrations, and experiments to engage students actively and visual aids like presentations, videos, and models to enhance understanding.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Assessment Types</b>               | <div data-bbox="432 521 1524 651"> <b>MODE OF ASSESSMENT</b><br/><br/> Mode of assessment: <b>Theory</b> </div> <div data-bbox="432 651 1524 1151"> <b>A. Continuous Comprehensive Assessment (CCA)</b><br/><br/> <b>Theory</b><br/> Total Marks – 30 Marks<br/> Assessment Methods <ul style="list-style-type: none"> <li>• Assignments</li> <li>• MCQ</li> <li>• Viva</li> <li>• Class tests</li> </ul> </div> <div data-bbox="432 1151 1524 2004"> <b>End Semester Evaluation (ESE)</b><br/> <b>Theory</b><br/> Total Marks – 70 Marks<br/> <b>Duration of Examination:</b> 2 Hours<br/><br/> <b>Pattern of examination for Theory:</b> Non - MCQ<br/> <b>Different parts of written examination:</b> Part A, Part B, Part C<br/><br/> <b>Answer Type:</b> Two sentence, short answer, essay<br/><br/> <b>Marks for part:</b> <ul style="list-style-type: none"> <li>• Part A<br/>Answer in two sentences - (10 out of 12) - <math>10 \times 2 = 20</math></li> <li>• Part B<br/>Short answer- (6 out of 8) - <math>6 \times 5 = 30</math></li> <li>• Part C<br/>Essay - (2 out of 4) - <math>2 \times 10 = 20</math></li> </ul> </div> |

## REFERENCES

- T. Coultate, Food: The Chemistry of Its Components, 6th Edn. RSC. 2015.
- S. Chowla, Engineering Chemistry, Danpat Rai Publication, 2020.
- B.K. Sharma. Industrial Chemistry, Krishna Prakashan, 2023.
- CNR Rao- Understanding chemistry, Universities Press, 1999.
- A. K. De, Environmental Chemistry, New age International Ltd. 2021.
- S. S. Dara, A Textbook of Environmental chemistry and pollution control, S.Chand & Company Ltd, 2004.
- Tisdale, S.L., Nelson, W.L. and Beaton, J. D. Soil Fertility and Fertilizers, Macmillian Publishing Company, New York, 1990.
- Buchel, K.H. Chemistry of Pesticides, John Wiley & Sons, New York, 1983.



**MGU-B.VOC. (HONOURS)**

## Syllabus

# SEMESTER 6



MGU-B.VOC. (HONOURS)

Syllabus



# Mahatma Gandhi University

## Kottayam

|                        |                                                                                                                                                        |         |           |     |             |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------|-----|-------------|
| Faculty/<br>Discipline | SCIENCE / CHEMISTRY                                                                                                                                    |         |           |     |             |
| Programme              |                                                                                                                                                        |         |           |     |             |
| Course Name            | ADVANCED PHYSICAL CHEMISTRY                                                                                                                            |         |           |     |             |
| Type of Course         | MPC                                                                                                                                                    |         |           |     |             |
| Course Code            | MG6MPCCHY300                                                                                                                                           |         |           |     |             |
| Course Level           | 300-399                                                                                                                                                |         |           |     |             |
| Course Summary         | Explore various branches of chemistry including thermodynamics, photochemistry, nanotechnology, spectroscopy, surface chemistry and chemical kinetics. |         |           |     |             |
| Semester               | 6                                                                                                                                                      | Credits |           | 4   | Total Hours |
| Course Details         | Learning Approach                                                                                                                                      | Lecture | Practical | OJT |             |
|                        |                                                                                                                                                        | 4       |           |     |             |
| Pre-requisites, if any | Nil                                                                                                                                                    |         |           |     |             |

### COURSE OUTCOMES (CO)

| CO No. | Expected Course Outcome                | Learning Domains | PO No |
|--------|----------------------------------------|------------------|-------|
| 1      | Interpret various thermodynamic terms. | U                | 1,2   |

|   |                                                                   |   |          |
|---|-------------------------------------------------------------------|---|----------|
| 2 | Describe laws of photochemistry and illustrate Jablonski diagram. | U | 1,2      |
| 3 | Understand the fundamentals of molecular spectroscopy.            | U | 1,2      |
| 4 | Explain the basics of nanotechnology.                             | U | 1,2,7,10 |
| 5 | Summarize the concepts of surface chemistry.                      | U | 1,2      |
| 6 | Explain the concept of activation energy and Arrhenius equation.  | U | 1,2      |

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I), Appreciate (Ap).**

#### CO-PO ARTICULATION MATRIX

| CO/PO | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO 10 |
|-------|------|------|------|------|------|------|------|------|------|-------|
| CO 1  | 2    | 3    |      |      |      |      |      |      |      |       |
| CO 2  | 2    | 3    |      |      |      |      |      |      |      |       |
| CO 3  | 2    | 3    |      |      |      |      | 2    |      |      | 2     |
| CO 4  | 2    | 3    |      |      |      |      |      |      |      |       |
| CO 5  | 2    | 3    |      |      |      |      |      |      |      |       |
| CO 6  | 2    | 3    |      |      |      |      |      |      |      |       |

**'0' is No Correlation, '1' is Slight Correlation (Low level), '2' is Moderate Correlation (Medium level) and '3' is Substantial Correlation (High level).**

#### COURSE CONTENT

##### Content for Classroom transaction (Units)

| Module | Units                                                    | Course description                                                                                                                   | Hrs | CO No. |
|--------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----|--------|
| 1      | <b>FUNDAMENTALS OF THERMODYNAMICS AND PHOTOCHEMISTRY</b> |                                                                                                                                      |     |        |
|        | 1.1                                                      | Thermodynamic terms- system, surroundings, types of system. Reversible and irreversible process, Extensive and intensive properties, | 2   | 1      |

|   |                                               |                                                                                                                                                                             |   |   |
|---|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
|   | 1.2                                           | Laws of thermodynamics (statement only), Enthalpy, Spontaneity, Entropy.                                                                                                    | 2 | 1 |
|   | 1.3                                           | Photochemistry- Introduction, Photochemical vs thermal reactions.                                                                                                           | 1 | 2 |
|   | 1.4                                           | Laws of photochemistry. Quantum yield,                                                                                                                                      | 2 | 2 |
|   | 1.5                                           | Photosensitization, Photosynthesis, Flash photolysis, Chemiluminescence.                                                                                                    | 3 | 2 |
|   | 1.6                                           | Jablonski diagram, Fluorescence and phosphorescence                                                                                                                         | 2 | 2 |
|   | <b>FUNDAMENTALS OF MOLECULAR SPECTROSCOPY</b> |                                                                                                                                                                             |   |   |
| 2 | 2.1                                           | Introduction: electromagnetic radiation, regions of the spectrum, interaction of electromagnetic radiation with matter, various types of molecular spectroscopy techniques. | 2 | 3 |
|   | 2.2                                           | Rotational spectroscopy: Introduction, principles and applications.                                                                                                         | 2 | 3 |
|   | 2.3                                           | Vibrational spectroscopy: Introduction, principles, applications, Raman spectroscopy (Elementary idea).                                                                     | 4 | 3 |
|   | 2.4                                           | Electronic spectroscopy: Introduction, principles and applications.                                                                                                         | 2 | 3 |
|   | 2.5                                           | Nuclear Magnetic Resonance(NMR) spectroscopy: Introduction, principles and applications.                                                                                    | 2 | 3 |
|   | 2.6                                           | Mass spectroscopy: Introduction, Principle, applications.                                                                                                                   | 3 | 3 |

|   |                                                |                                                                                                                                                                                  |   |   |
|---|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
| 3 | <b>FUNDAMENTALS OF NANOTECHNOLOGY</b>          |                                                                                                                                                                                  |   |   |
|   | 3.1                                            | History of nanomaterials ,Different types of nanomaterials, classification of nanomaterials based on dimensions and based on origin.                                             | 3 | 4 |
|   | 3.2                                            | Bottom up approach and top down approach.                                                                                                                                        | 2 | 4 |
|   | 3.3                                            | Preparation, properties and application of nanomaterials- carbon nanotubes, quantum dots.                                                                                        | 4 | 4 |
|   | 3.4                                            | Application of nanotechnology in the food industry- Food packaging, food processing and preservation, food sensors, food safety and quality control.                             | 4 | 4 |
| 4 | <b>SURFACE CHEMISTRY AND CHEMICAL KINETICS</b> |                                                                                                                                                                                  |   |   |
|   | 4.1                                            | Adsorption- Types of adsorption of gases by solids, factors influencing adsorption, Freundlich Adsorption isotherm, Langmuir Adsorption isotherm (derivation not required).      | 3 | 5 |
|   | 4.2                                            | Colloids-preparation, properties- optical and electrical, Electrical double layer, coagulation, electrophoresis, electro osmosis, surfactants, micelle, application of colloids. | 4 | 5 |
|   | 4.3                                            | Rate of reaction, rate equation, order and molecularity of reactions, determination of order of a reaction.                                                                      | 4 | 6 |
|   | 4.4                                            | Effect of temperature on the rate of reaction: Arrhenius equation, concept of activation energy, importance of activated                                                         | 4 | 6 |



|   |                                 |                                                                                                                                                                                                      |   |   |
|---|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|
|   |                                 | complex.Collision theory,<br>Transition state theory.                                                                                                                                                |   |   |
|   | 4.5                             | Catalysis: Homogeneous catalysis, enzyme catalysis – Michaelis – Menten equation (no derivation needed). Heterogeneous catalysis – Surface catalysis, Elementary idea about Autocatalysis. Examples. | 5 | 6 |
| 5 | <b>Teacher Specific Content</b> |                                                                                                                                                                                                      |   |   |
|   |                                 |                                                                                                                                                                                                      |   |   |

|                                       |                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Teaching and Learning Approach</b> | <b>Classroom Procedure (Mode of transaction)</b><br><br>Lecture Sessions, interactive sessions including discussions, demonstrations, and experiments to engage students actively and visual aids like presentations, videos, and models to enhance understanding.                                                       |
| <b>Assessment Types</b>               | <b>MODE OF ASSESSMENT</b><br>Mode of assessment: <b>Theory</b><br><br><b>A. Continuous Comprehensive Assessment (CCA)</b><br><br><b>Theory</b><br>Total Marks – 30 Marks<br><br>Assessment Methods <ul style="list-style-type: none"> <li>• Assignments</li> <li>• MCQ</li> <li>• Viva</li> <li>• Class tests</li> </ul> |
|                                       | <b>End Semester Evaluation (ESE)</b><br><br><b>Theory</b><br>Total Marks – 70 Marks                                                                                                                                                                                                                                      |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>Duration of Examination:</b> 2 Hours</p> <p><b>Pattern of examination for Theory:</b> Non - MCQ</p> <p><b>Different parts of written examination:</b> Part A, Part B, Part C</p> <p><b>Answer Type:</b> Two sentence, short answer, essay</p> <p><b>Marks for part:</b></p> <ul style="list-style-type: none"> <li>• Part A<br/>Answer in two sentences - (10 out of 12) - <math>10 \times 2 = 20</math></li> <li>• Part B<br/>Short answer- (6 out of 8) - <math>6 \times 5 = 30</math></li> <li>• Part C<br/>Essay - (2 out of 4) - <math>2 \times 10 = 20</math></li> </ul> |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## REFERENCES

- N. Kumar, K. Sunita, Essentials in Nanoscience and Nanotechnology, Wiley, 2016.
- Pradeep, T. NANO: The Essentials: Understanding Nanoscience and Nanotechnology; 1st Edition ed.; McGraw-Hill Education: New York, 2007.
- Muralidharan, V. S.; Subramania, A. Nanoscience and Technology; Ane Books for, 2009.
- Puri, Sharma, Pathania, Principles of Physical Chemistry, 48th Edn. Vishal Publishing Company, 2012
- F. Daniels and R. A. Alberty, Physical Chemistry, 3rd ed., John Wiley and Sons, Inc., New York, 2021.
- G.M. Barrow, Physical Chemistry, Tata McGraw-Hill, 2007.
- G. W. Castellan, Physical Chemistry, 4th Edn. Narosa Publishing House, 2018.
- K. L. Kapoor, A Textbook of Physical chemistry, Volume 5, 4th edition, Macmillan India Ltd., 2018.
- D. A. McQuarrie, J. D. Simon, Physical Chemistry – A molecular Approach, Viva Books Pvt. Ltd., 2019.
- G. Raj, Advanced Physical Chemistry, Goel Publishing House, 2016.